

Robust Learning-Based Camera Motion Characterization Scheme With Applications to Video Stabilization

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Abstract—This paper investigates a novel learning-based camera motion characterization scheme along with its application to the video stabilization problem. The proposed characterization scheme represents the compressed domain block motion vectors (MVs) using polar angle and magnitude histograms. Discriminative features from these two histograms are extracted and fed to a supervised learning-based hierarchical classifier for recognizing the six camera motion patterns. A comparative analysis with an existing scheme is carried out to support and validate the proposed characterization scheme. The proposed scheme works at the frame level by classifying the inter-frame camera motion patterns. This scheme is extended to classify the video segments and a novel application to video stabilization is investigated. An experimental analysis of a number of test sequences captured using a handheld video camera shows that by characterizing the smooth and jittery motions, selective video stabilization could be carried out only on those video segments that have been degraded. This approach of selective video stabilization saves considerable amount of computational time compared with running the stabilization algorithm on the entire video sequence, as proposed in the literature. The proposed strategy of using a classification scheme prior to applying the video stabilization routine offers a new paradigm to the conventional video stabilization problem. Experimental validation carried out using exhaustive search motion estimation obtained block MVs, and H.264/Advanced Video Coding-obtained MVs shows that by using the idea of selective video stabilization, up to 62% reduction in processing time can be achieved compared with video stabilization approaches wherein the entire video sequence is processed.

Index Terms—Backpropagation neural network, block motion vectors (MVs), camera motion, hierarchical classification, supervised learning, video stabilization.

I. INTRODUCTION

CAMERA motion and object motion are two main sources of dynamic information in videos. Camera motion, also referred to as the global motion, comprises of pan, tilt, zoom, and/or a combination of these basic components. On the other hand, object motion also referred to as local motion

is caused due to the movement of objects in a scene. Knowledge of the camera motion is useful in applications like video stabilization, moving object segmentation, video shot segmentation, video coding, super resolution, structure from motion, video indexing, error concealment, and so on. Video stabilization aims at aligning video frames that have undergone misalignment due to annoying hand motions and/or platform vibrations. Typically, three major stages constitute a video stabilization pipeline: 1) camera motion estimation; 2) motion smoothing; and 3) motion compensation. Camera motion estimation is an important step in the stabilization pipeline and its accuracy determines the stabilization performance.

In [1], camera motion was estimated from local motions of four subimages which was detected using peak values of phase correlation surfaces and its application was explored for video stabilization. Ko *et al.* [2] explored a fast video stabilization technique utilizing a computationally efficient gray coded bit-plane matching technique to estimate the local motion vectors (MVs) followed by using the local estimates for estimating the global motion. Battiato *et al.* [3] utilized a set of simple rejection rules to estimate the inter-frame camera transformation parameters from local MVs and explored its application for video stabilization. They further extended this paper by adopting their method for mobile imaging devices and tested it on ARM device achieving real time performance [4]. Puglisi and Battiato [5] explored a novel technique for video stabilization by utilizing an exhaustive search strategy with an integral projection-based error function for obtaining the block-based local MVs followed by computing the global inter-frame motion parameters through a voting-based method. Yang *et al.* [6] utilized correspondences between SIFT points to obtain a rough estimate of the projected camera motion followed by refining this estimate using particle filtering strategy with very good results. Weng and Jiang [7] proposed a fast camera motion estimation technique utilizing the closed form mathematical relationship between camera motion parameters, corresponding points between neighboring frames and the MPEG MVs. A full-frame video stabilization method was proposed in [8], where a combination of motion in painting and image deblurring provided an effective enhancement on the quality of stabilized image sequences. Okade and Biswas [9] proposed a novel compressed domain framework for video stabilization that was fast and robust in comparison to the state-of-the-art pixel-based stabilization techniques. Okade and Biswas [10] utilized wavelet analysis to estimate the camera motion parameters from block MVs and showed

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that it offered computational advantages over existing methods. Su *et al.* [11] proposed a novel and robust global motion estimation technique that utilized Newton–Raphson gradient descent optimization technique with outlier handling to minimize the difference between the estimated and actual MVs using error minimization. Chang *et al.* [12] used optical flow to estimate the local MV field followed by estimating the translational and rotational motions using a least square estimation technique. In [13], a video stabilization technique for a camera mounted on a moving vehicle was explored by directly calculating the 3-D camera motions using lane lines as global features instead of the traditional 2-D inter-frame motion calculations. A robust and efficient 2-D affine global motion estimation algorithm based on phase correlation in the Fourier–Mellin domain with least square model fitting of sparse MVs and its application for digital image stabilization was explored in [14]. Liu *et al.* [15] placed subspace constraints on feature trajectories by tracking the feature points followed by computing the locations of the tracked points and finally rendered an output video that followed the feature point locations. Camera motion characterization [16]–[25] aims at finding a robust representation scheme for the camera motion, so that features extracted from this representation will aid in recognizing the inter-frame camera motion types. In [17], a six-parameter transformation model was utilized to classify camera motions into panning, tilting, and zooming and its application to annotation of basketball videos was investigated. Their method was purely based on using the MPEG compressed domain motion information to carry out the classification task. A camera motion classification and key-frame extraction strategy that operates in the block MV domain was proposed in [18], and its applications were explored for efficient video content accessing. Recently, Kowdle and Chen [19] have used a learning mechanism to segment a scene into static and dynamic parts using discrete MRF labeling via graph cut in the context of time stamping archival movies. A robust and accurate motion annotation engine incorporating novel zoom detection and motion magnitude profiling algorithms entirely using the MPEG-2 compressed video was proposed in [20]. However, their strategy was sensitive to outliers in the MVs. Duan *et al.* [16] investigated a novel nonparametric scheme for characterizing the camera motion using a learning-based technique with the objective of utilizing it for indexing and retrieval applications. Their method utilized mean shift clustering on block MVs to obtain the dominant motion mode corresponding to a particular camera motion type with very good results. Camera motion characterization has traditionally found its application in indexing and retrieval, while in this paper the motivation is to bring its application for an important video processing task like stabilization.

The contributions in this paper are twofold. First, we investigate a novel and robust camera motion characterization scheme utilizing the mutual relationship shown by the compressed domain block MVs. The proposed scheme is nonparametric as no explicit motion model is assumed and it operates at the frame level by identifying the inter-frame camera motion type. Learning-based hierarchical

classification is used for identifying the six camera motion patterns. Second, the proposed characterization scheme is utilized in identifying the jittery portions of the video sequence, so that selective stabilization can be carried out only on the degraded video segments thereby offering a new paradigm to the video stabilization problem. By our extensive experiments, we show and analyze how selective stabilization using a classification strategy saves considerable amount of computational time in comparison with techniques which run the stabilization routine over the entire video sequence. The research results and dataset used in this paper are available at <http://www.facweb.iitkgp.ernet.in/~pkb/>. The motivation for making the research results public is to allow it to be reproducible and useful for fellow researchers.

II. PROPOSED CAMERA MOTION CHARACTERIZATION FRAMEWORK

The proposed framework is based on utilizing the block MVs available in the compressed bitstream for carrying out the inter-frame camera motion pattern classification. The strategy is basically decoder driven as it utilizes the block MVs resulting from the video encoding process. Novel and discriminative features are extracted from the block MVs to distinguish the camera motion types. These features are fed to a supervised learning-based hierarchical classification scheme that generates the classification labels corresponding to the recognized camera motion pattern.

A. Camera Motion Patterns

The focus in this paper is to robustly classify the following inter-frame camera motion patterns using the block MVs available in the compressed bitstream: zooming camera (ZC), panning right camera (PRC), panning left camera (PLC), tilting up camera (TUC), tilting down camera (TDC), and still camera (SC).

B. Motivation for Using Block Motion Vectors

The reason for the choice of block MVs as an input to the system is twofold. First, it is generic and can be modified to any codec, i.e., H.264 or MPEG-4. Second, such input data could come from other sources like feature correspondence, tracking, or even full optic flow itself. The modification and adoption to H.264 codec is shown as an example case study in Section VI.

C. Proposed Scheme

Fig. 1 shows the framework of the proposed camera motion characterization scheme. The framework is based on representing the block MVs using polar magnitude and angle histograms, so that statistical discriminative features drawn from these histograms will aid in the classification task. The features are fed to a supervised learning-based hierarchical classifier which classifies the camera motion patterns into classification labels, as listed in Section II-A. The detailed description of the proposed framework is given as under.

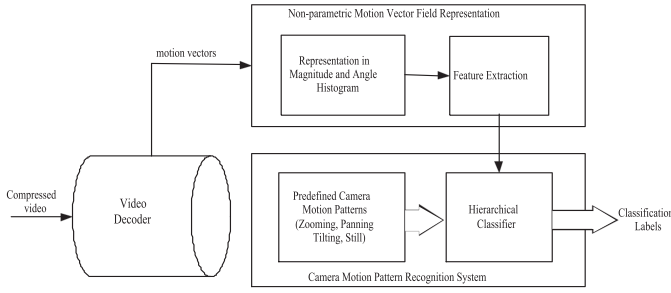


Fig. 1. Proposed camera motion characterization framework.

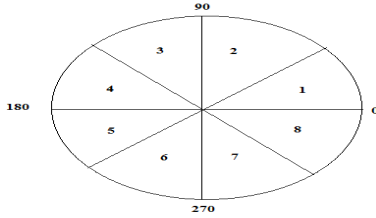


Fig. 2. Polar angle histogram.

Let MV^X and MV^Y denote the horizontal (X) and vertical (Y) component of a block MV , respectively, i.e., the vector $MV = (MV^X, MV^Y)$. The block MVs are transformed from the Cartesian to polar representation as follows:

$$MV_{\text{mag}} = \sqrt{(MV^X)^2 + (MV^Y)^2} \quad (1)$$

$$MV_{\text{ori}} = \arctan\left(\frac{MV^Y}{MV^X}\right) \quad (2)$$

where MV_{mag} gives the magnitude of the block MVs , while MV_{ori} gives its orientation. The polar angle histogram (as shown in Fig. 2) is obtained using the orientation of the block MVs (2). It shows the distribution of angle values according to their numeric range in angle bins. The length of each bin reflects the number of MVs that fall within a group which ranges from zero to the greatest number of block MVs deposited in that bin. The magnitude histogram is obtained using (1). The proposed classification scheme is mainly based on novel and simple statistical features extracted from polar angle and magnitude histograms.

1) Features From Polar Angle and Magnitude Histograms:

The polar angle histogram is shown in Fig. 2. The MVs in the angle histogram are grouped into eight sectors ($s_i, i = 1 \dots 8$) for analysis purpose. The sectors are marked anticlockwise, as shown in Fig. 2. Table I shows the polar angle and magnitude histograms for camera motion types: ZC, PLC, TUC, TDC, SC, and PRC. The video sequences, from which these histograms are generated for the respective patterns, are listed in Table III. The following observations are made from the polar angle histogram (MV_{ori}) and the magnitude histogram (MV_{mag}) for the six known camera motion patterns.

ZC: Roughly an even spread of MVs in all eight sectors of the angle histogram for a ZC case.

PLC: The dominant angle bins being located in Sector 4 and/or Sector 5 of the angle histogram for PLC case.

TUC: The dominant angle bins being located in Sector 2 and/or Sector 3 of the angle histogram for TUC case.

TDC: The dominant angle bins being located in Sector 6 and/or Sector 7 of the angle histogram for TDC case.

SC: A single dominant bin for SC case. The dominant angle bin is located in Sector 1 of the angle histogram. The magnitude histogram in this case shows a dominant peak at the first bin location signifying that the camera has not moved.

PRC: The dominant angle bins are again located in Sector 1 and/or Sector 8 of the angle histogram. However, the magnitude histogram in this case does not show a dominant peak at the first bin location due to the camera having moved resulting in finite magnitude for the MVs .

The sequences used had object motion present in them. Object motion is represented by tiny bins in the angle histogram and does not affect the classification as long as the object does not cover more than 60% of frame size. For cases where moving objects cover more than 60% of the frame area (i.e., object motions dominating the camera motions), our experiments show a limiting case as analyzed with an challenging test sequence in Section VI. (Please refer to Section VI-G.)

Based on the above observations the following inferences are drawn.

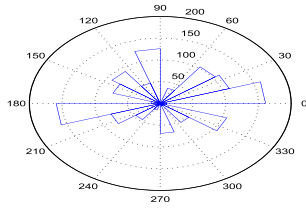
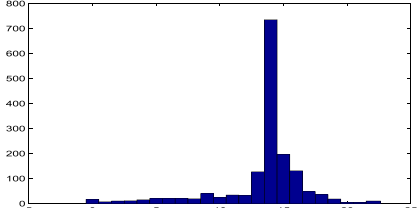
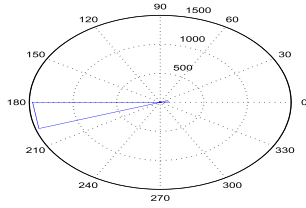
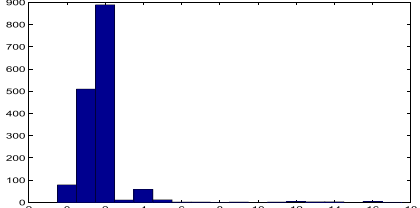
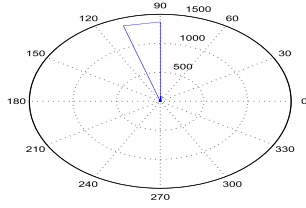
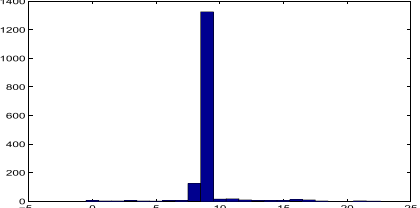
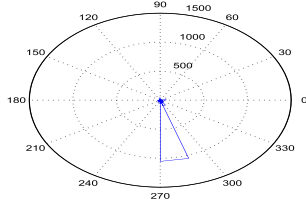
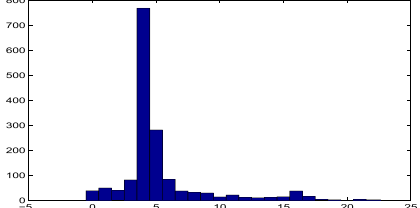
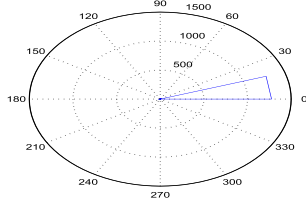
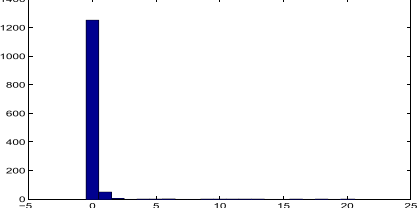
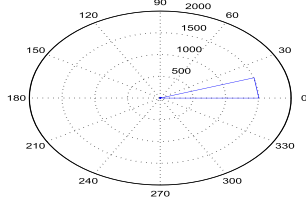
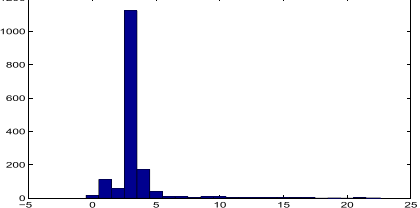
- 1) Even spread of bins in all sectors of the angle histogram for a ZC pattern.
- 2) Dominant bins in the angle histogram for rest of the camera motion patterns.
- 3) Dominant bins location in the angle histogram signifying the direction of camera movement.
- 4) Dominant peak at the first bin location of the magnitude histogram for an SC pattern.

Statistical features validating the above inferences are extracted for classification of the camera motion patterns as follows. At the first level, a ZC case can be separated utilizing the fact that it has an even spread of MVs across all sectors. A measure based on variance can be used for this purpose. The coefficient of variation between the sectors (cv_s) is one such novel and discriminating feature which is utilized in the proposed scheme for classifying the ZC case and is given by

$$cv_s = \frac{\sigma_s}{\mu_s} \quad (3)$$

where σ_s and μ_s are the standard deviation and mean of the eight sector counts, respectively. A small value of coefficient of variation signifies that the MVs are evenly distributed in all the sectors which occur in case of a ZC, whereas a very large value signifies that the spread is uneven which may be due to the camera panning, tilting, or remaining still. Based on this reasoning, the cv_s is used as a discriminative feature for broadly separating the patterns into ZC and others. At the second level, the others (i.e., remaining camera patterns) contain the following patterns that need separation: PLC, TUC, TDC, PRC, and SC. The dominant bin location in the polar angle histogram is utilized as a discriminating feature to make this decision. Three classes are clearly recognized: PLC, TUC, and TDC. However, as observed the angle histogram cannot be

TABLE I
POLAR ANGLE AND MAGNITUDE HISTOGRAMS FOR SIX CAMERA MOTION PATTERNS

Pattern	Angle histogram	Magnitude histogram
ZC		
PLC		
TUC		
TDC		
SC		
PRC		

used to distinguish PRC and SC as the dominant bin may be located in the same sector in both cases. This is because PRC has majority of MVs with orientation range of 315° – 45° , while

an SC has majority of MVs with orientation range of 0° – 10° in the angle histogram. This may lead to the dominant bin being located in Sector 1 which would confuse between

TABLE II
FEATURE CHARACTERISTICS FOR CAMERA MOTION PATTERNS

Camera Pattern	cv_s	Angle histogram Dominant bins	Magnitude histogram first bin
ZC	Small	Spread across sectors	No Dominant Peak
PLC	Large	Sector 4 and/or Sector 5	No Dominant Peak
TUC	Large	Sector 2 and/or Sector 3	No Dominant Peak
TDC	Large	Sector 6 and/or Sector 7	No Dominant Peak
PRC	Large	Sector 1 and/or Sector 8	No Dominant Peak
SC	Large	Sector 1	Dominant Peak

an SC and a PRC. Hence, at the third level, the magnitude histogram is utilized to break this tie as for an SC case the magnitude histogram contains a dominant peak at the first bin location. The first bin in the magnitude histogram corresponds to zero motion (i.e., no motion) and is in majority only when the camera is still where as for a PRC the histogram bins have finite magnitude depending on the extent of panning motion. Using the three features $\{cv_s, \text{dominant_angle_bin_placement}, \text{count_magnitude_histogram_bin_1}\}$, all the six camera motion patterns are distinctly recognized. Table II summarizes the feature characteristics for the six camera motion types.

2) *Supervised Learning Using Neural Networks*: The idea is to learn the features important to classification as part of learning a classifier. Supervised learning is adopted in this paper to learn the decision boundary at different levels of the decision making process. As described in the previous section, three features are utilized for classifying the camera patterns. The values of these features for class separation are learnt from the training data as part of supervised learning process. The video sequences used for the learning are listed in Table III. Three neural network classifiers nn_1 , nn_2 , and nn_3 are trained using backpropagation algorithm to obtain the mappings w_1 , w_2 , and w_3 , respectively. w_1 uses coefficient of variation (cv_s) as a feature for learning, w_2 uses dominant angle bin placement as a feature for learning, while w_3 uses count of the first bin in the magnitude histogram as a feature for learning. Normalization is performed with respect to the frame size ($W \times H$) for the magnitude histogram, so that the first bin of the histogram can be used irrespective of the frame resolution of the video.

3) *Hierarchical Classification*: Since the classification task is multiclass, hierarchical classification is used. The basic idea involved in any multiclass problem is to break up a complex decision into a union of several simple decisions. This concept is exploited in the proposed scheme by classifying the camera patterns hierarchically. Three levels, i.e., L_1 , L_2 , and L_3 are used for this purpose. At each level, a single feature is used in a supervised manner for recognizing the camera motion patterns. Fig. 3 shows the block diagram for hierarchical classification with six predefined camera motion patterns. The classification is carried out using the three features, i.e., $\{cv_s, \text{dominant_angle_bin_placement}, \text{count_magnitude_histogram_bin_1}\}$. At the first level, i.e., Level 1 (L_1), $\{cv_s\}$ is used as a feature to broadly classify the ZC pattern. At Level 2 (L_2),

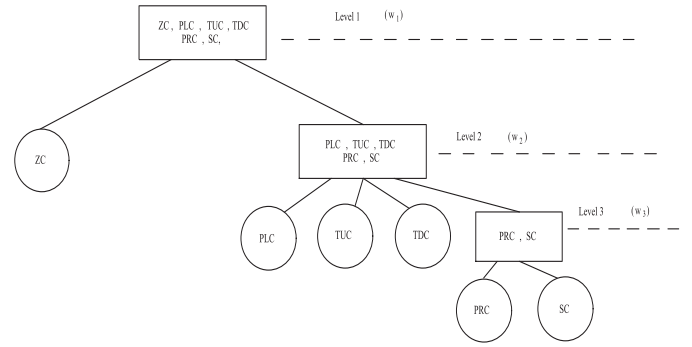


Fig. 3. Hierarchical classification using neural networks.

the $\{\text{dominant_angle_bin_placement}\}$ is used as a feature for classifying PLC, TUC, and TDC patterns. At Level 3 (L_3), the $\{\text{count_magnitude_histogram_bin_1}\}$ is used as a feature for classifying SC and PRC patterns. Advantage with hierarchical classification is that at each level we can evaluate an attribute from the data by a comparison test and learn the decision boundary to arrive at the next level which may be the classification label or a union of classes where again a comparison test is needed. This process continues until we arrive at individual classification labels. This process can be visualized from Fig. 3, where three levels are utilized for classifying the six camera motion patterns using a single feature at each level. The classifier mappings (w_1 , w_2 , and w_3) for class separation at each level are obtained via supervised learning from the training data, as described in the previous section. L_1 uses w_1 , L_2 uses w_2 , and L_3 uses w_3 in the proposed scheme. In our case, the decision boundary selection for class separation is performed by training neural network classifiers using features extracted from the training sequences thereby bringing in automatic boundary separation. Researchers have tried modeling this multiclass classification task using binary classifiers such as the support vector machine (SVM) [16], [26]. However, the drawback in using SVM is that since it is essentially a two class classifier, so for multiclass adoption we would need it to repeat several times, i.e., for each class versus others class thereby increasing the number of classifiers to be trained. Since we had a multiclass requirement at L_2 , using a single neural network, i.e., nn_2 , good class separation is achieved in our case.

III. APPLICATION TO VIDEO STABILIZATION

A. Motivation

The motivation behind the design of the proposed classification scheme is to use it in the context of video stabilization. Currently, the focus in the video processing community is on designing new and better schemes for video stabilization and not on using intelligence as to when the stabilization routine should be invoked. In this paper, we take a paradigm shift from this conventional approach and propose to use camera motion classification prior to the stabilization stage due to the following motivation. In a handheld camera, jitters and/or environment conditions might degrade the quality of video thereby warranting the need for stabilization algorithms.

TABLE III
DATASET USED FOR TRAINING AND TESTING

Sequence Names	Available at	Training Sequences	Testing Sequences
Tractor, Station2, Shields, Stockholm, Parkrun	https://cs-nsf-wiki.cs.surrey.sfu.ca/wiki/Video_Library_and_Tools	Diving, Tractor, Station2, Shields, Stockholm, Jets, Parkrun, Biathlon Penguins, Allstars, Coastguard, Snow leopard & mountain goat	Stefan & Desert
Penguins, Jets, Desert, Allstars, Biathlon	http://ftlabsrv.nue.tu-berlin.de/review/sequences/		
Stefan, Coastguard, Table tennis	http://media.xiph.org/video/derf/		
Diving, Snow leopard and mountain goat	http://www.facweb.iitkgp.ernet.in/~pkb/		

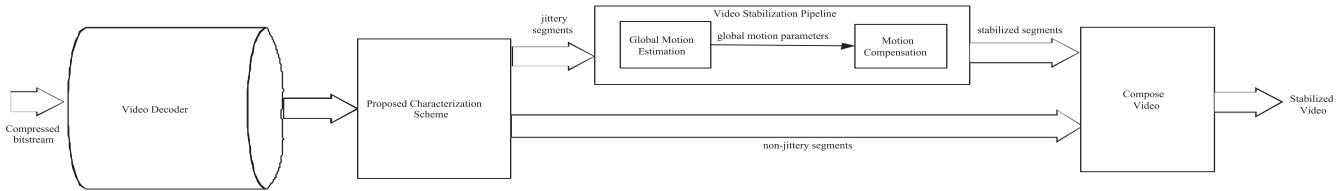


Fig. 4. Novel application of the proposed characterization scheme to video stabilization.

However, these unwanted motions are not present throughout the video and may appear during the course of video capture. This indicates that in many cases only portions of the video are degraded. As a consequence of this observation there is a need to identify the video segments which require stabilization. By bringing such knowledge, considerable amount of processing time can be reduced as only selective video segments can be stabilized unlike the entire sequence as done presently. The proposed scheme can also be useful in automatic detection of degraded portions in a video sequence. Fig. 4 shows the novel application of the proposed characterization scheme to video stabilization. The idea is to first identify the degraded video segments and apply the stabilization selectively on those segments to obtain the stabilized sequence. The selection of the degraded segments is carried out using the proposed characterization scheme as follows.

B. Identifying Jittery Video Segments

A label change is defined as the change in the behavior of the camera, e.g., zooming to panning, panning to tilting, tilting to zooming, and so on. The desired camera motions are generally characterized by smooth label changes where as a camera undergoing unsteady motion will have frequent label changes during a video segment. In other words, the transition between labels would be less frequent for a desired video segment, while the undesired video segment would be experiencing frequent change in label types due to the unsteadiness of the video camera. The number of such label changes during a video segment is counted and compared against a threshold. If the label change count exceeds the threshold then the video segment is classified as undesired else it is classified as desired. The need for such fine grained categorization (i.e., six labels) for the camera is felt,

so that the camera's usual desired motions like pan left, pan right, tilt up, tilt down, zoom, can still be identified and be differentiated from the cameras haphazard motion, where the listed motion types make a rapid transition thereby indicating the camera may be experiencing turbulence (jitters). In some cases when motion changes occur through a combination of such involved classes, the proposed method would consistently output labels that belong to either of the involved classes. For example, when there is simultaneous zooming along with panning, it is a case of combination of two involved classes, i.e., zoom and pan. In such a scenario the classifier outputs the labels as either zoom or pan, so that the labels will be consistent with the camera motion that is taking place.

The identified undesired segments are then stabilized using our stabilization algorithm [9] for compressed videos whose summarized description appears in Section III-C. The pseudocode of the novel application of the proposed characterization scheme for video stabilization is given above. Lines 1–11 of Algorithm 1 identify the video segments which need to be processed by the stabilization routine, while Lines 13–17 of Algorithm 1 perform the stabilization task on the selected segments by estimating the camera motion parameter (m) followed by smoothing the transformation chain and building the compensated frames.

C. Summarized Description of Video Stabilization Pipeline

In this section, we present the summarized description of the video stabilization pipeline shown in Fig. 4. Readers are kindly referred to [9] for details on each of the mentioned steps as well as comparative evaluations with state-of-the-art pixel domain stabilization methods like PFME method [6] and subspace method [15]. As observed from Fig. 4, once the jittery segments are identified by the proposed characterization

Algorithm 1 Novel Application of Camera Motion Classification to Video Stabilization

Input: Block MVs extracted from compressed bitstream.

Output: Stabilized video.

1. video segment size (s_f) = video frame rate.
 2. N_f = Total number of frames in video
 3. $\text{total_segments} = \frac{N_f}{s_f}$.
 4. **for** $j = 1$ to total_segments **do**
 5. Label all frames of segment j using the proposed classification scheme of Section II
 6. **if** (label change count) $_j >$ Threshold **then**
 7. Mark video segment as undesired and add to Queue.
 8. **else**
 9. Mark video segment as desired.
 10. **end if**
 11. **end for**
 12. Perform video stabilization only on undesired video segments as follows
 13. **while** (Queue not empty) **do**
 14. Extract segment from queue and estimate the global motion parameters using

$$\mathbf{m}_t = \arg \min_{\mathbf{m}} \sum_{(x,y)} \|\mathbf{MV}(x, y, t) - \mathbf{MV}(x, y; \mathbf{m})\|^2.$$
 15. Smooth the global motion parameters to retain the desired motions and finally warp the video frame to construct the motion compensated video frame using $I_t \leftarrow I_t'(S_t)$.
 16. Store the stabilized segment in the compose block
 17. **end while**
 18. Join the desired segments and the stabilized segments to obtain the final video.
-

scheme as described earlier, these segments are stabilized as follows.

- 1) The global motion parameters between successive frames are estimated using the compressed domain MVs utilizing Su *et al.*'s [11] method which minimizes the following squared error term:

$$\mathbf{m}_t = \arg \min_{\mathbf{m}} \sum_{(x,y)} \|\mathbf{MV}(x, y, t) - \mathbf{MV}(x, y; \mathbf{m})\|^2 \quad (4)$$

where $\mathbf{MV}(x, y; \mathbf{m})$ is the MV which is estimated using the horizontal (X -) and vertical (Y -) component of a MV, i.e., $\mathbf{MV}^X(x, y; \mathbf{m}) = x' - x$ and $\mathbf{MV}^Y(x, y; \mathbf{m}) = y' - y$. Here, (x, y) is the point location in the reference frame and (x', y') is the point location after the frame has undergone perspective transformation parameterized by $\mathbf{m} = [m_0, \dots, m_7]$. $\mathbf{MV}(x, y, t)$ is the actual MV at that location.

- 2) Once the global motion parameters are estimated they are smoothed to retain the desired motions, and finally the video frame is warped to construct the motion compensated video frame.
- 3) Finally, the stabilized segments and the nonjittery segments are composed to generate the stabilized video.

IV. ENFORCING TEMPORAL SMOOTHNESS

While the proposed characterization scheme achieves per frame classification, in some applications it may be necessary to incorporate temporal smoothness to ensure that the labeling which is obtained by the proposed characterization scheme does not keep switching too often between the label types. For the application of the proposed classification scheme to stabilization, we have grouped the frames into segments. As described earlier for every segment, we find the number of label changes (i.e., label change count) and those segments where this count exceeds the threshold are identified for processing by the stabilization routine. However, it might happen that the inter-frame labeling that is obtained keeps switching too often between the label types. To avoid this, we incorporate temporal smoothness using a moving average filter which smoothes the short-term fluctuations between the label types using

$$\widehat{CL} = \frac{1}{2k+1} \sum_{j=i-k}^{i+k} CL_j \quad (5)$$

where $\widehat{CL}$ is the smoothed label change count, CL_j is the camera label change count for segment number j , i is the frame index, k is the window size which is a design parameter that controls the amount of temporal smoothness needed, and $2k+1$ is the filter size. When $k=0$ there is no temporal smoothness enforced. This strategy of enforcing temporal smoothness is less sensitive to short-term noise and at the same time adaptive to long-term trends of the label types thereby adding stability and consistency to the proposed characterization scheme as opposed to not having any temporal smoothness enforced. Section VI-G1 shows the application of enforcing temporal smoothness on an example test sequence.

V. ANALYSIS OF COMPUTATIONAL COMPLEXITY

In this section, theoretical complexity analysis for the proposed characterization scheme as well as for its application to video stabilization is carried out. The following notations are used.

N_f	Total number of frames contained in the input video.
Q	Number of unstable video segments identified by the proposed characterization scheme.
s_f	Number of frames per segment.
n	Time taken per frame for stabilization.
m	Time taken per frame for classification.
$(Q * s_f / N_f)$	Normalized portion of the video that needs to be sent to the stabilization block.

Therefore, percentage of video on which stabilization is not needed is given by $(F_{\text{sav}}) = (1 - (Q * s_f / N_f)) * 100$. Since stabilization is taking n time units per frame, the possible time saving in time units $(T_{\text{sav}}) = (1 - (Q * s_f / N_f)) * n$.

Best case occurs when $Q * s_f = 0$, signifying none of the segments are unstable and maximum saving is achieved as the stabilizer module need not be run.

Worst case occurs when $Q * s_f = N_f$, signifying all segments are unstable and the proposed characterization scheme would have no role thereby posing as an additional overhead. The overhead analysis is carried out in Section V-A.

A. Overhead Analysis

Stabilization time n consists of time taken by GME block that involves error minimization, and GMC block that involves frame warping. Classification time m consists of time taken for feature extraction and mapping evaluation. Since the classifiers are trained offline, the classification time does not account for it in the computational analysis. In general, $n \gg m$ due to the GME and GMC steps consuming more time in comparison with feature extraction and mapping evaluations.

- 1) Stabilization complexity without the proposed characterization scheme = $N_f * n$.
- 2) Complexity of labeling frames using proposed characterization scheme = $N_f * m$.
- 3) Complexity with proposed characterization scheme integrated with the stabilizer = $N_f * m + Q * s_f * n$.

Classification will not be an overhead if the following condition holds:

$$\begin{aligned}
 N_f * m + Q * s_f * n &\leq N_f * n \\
 N_f * m &\leq (N_f - Q * s_f) * n \\
 m &\leq \left(\frac{N_f - Q * s_f}{N_f} \right) * n \\
 m &\leq F_{\text{sav}} * n \\
 F_{\text{sav}}(\%) &\geq \left(\frac{m}{n} \right) * 100
 \end{aligned}$$

i.e., if $(m/n) * 100\%$ frames being nonjittery (signifying they do not require stabilization) would ensure the proposed characterization scheme will not pose an overhead. For example, in our case for frame resolution 1280×720 , stabilization time n is 1.12 s, while classification time m is 0.113 s, so their ratio evaluates to 0.1008, i.e., $F_{\text{sav}} \geq 10\%$ signifying, the proposed application will not be overhead when nonjittery frames are at least more than 10%.

VI. RESULTS

MATLAB 7.10 is used on a Pentium dual core CPU running at 3 GHz with 2-GB RAM for experimentation. The block MV field is generated by exhaustive search motion estimation (ESME) algorithm considering block size = 8×8 , search_range = $[-8 \ 8]$, and cost_function set to MAD. An example case study using H.264/Advanced Video Coding (AVC) is also carried out to support and validate the proposed characterization scheme for real codecs. Please refer to Section VI-A. The input data could come from other sources like feature correspondence, tracking, or even full optic flow, where again the proposed characterization scheme would be applicable. Rose plot function in MATLAB is used to obtain the polar angle histogram. The angle histogram and magnitude histograms have 24 bins each. The pattern recognition toolbox PRTools developed by Delft University of Technology is used for carrying out the training and testing of the classifiers used



Fig. 5. Sample frames from the dataset used for training and testing. First row (left to right): *Allstars*, *Diving*, *Jets*, *Keiba*, and *Stockholm*. Second row (left to right): *Penguins*, *Station*, *Stefan*, *Tractor*, and *Parkrun*. Third row (left to right): *Desert*, *Mountain goat*, *Coastguard*, *Biathlon*, and *Table tennis*.

in this paper. The toolbox is available at <http://prtools.org/> and is free for academic research. The dataset used for evaluating the proposed scheme is diverse and consists of the camera motion patterns described in Section II-A. This dataset is taken from various sources listed in Table III. The *Diving* sequence is taken from Olympic diving swimming event as it contains predominantly tilting up and tilting down motions along with zooming of the camera following the diver which serves as the object. The *Snow leopard* sequence is taken from *Snow leopard* versus *Mountain goat* series and consists of predominantly panning right and tilting down motions of the camera along with object motions of the leopard chasing the goat. Sample frames from the mentioned sequences are shown in Fig. 5. The *Diving* and *Snow leopard* sequences are made available on our webpage listed in Table III. As mentioned earlier, all the tested sequences consists of object motion along with the camera motion. The proposed characterization scheme is compared with mean shift filtering based nonparametric classification scheme proposed in [16] as their method also utilizes block MVs in a learning-based framework. Their method is reimplemented on the above dataset to classify six camera motion patterns, so as to bring uniformity in comparison with the proposed characterization scheme.

A. H.264/AVC Case Study

An example case study for applicability of the proposed characterization scheme to real codecs is carried out by choosing H.264/AVC. The sequences, as listed in Table III, are encoded using the JM version 18 [27] encoder at different bitrates with GOP structure IPPP..., i.e., the first frame is intra (*I*) coded and subsequent frames are predictively (*P*) coded. Block MVs extracted from these encoded sequences serve as input to the proposed characterization scheme. Since H.264 defines four basic modes, i.e., 16×16 , 16×8 , 8×16 , and 8×8 , to have a uniformly sampled MV field, all the MVs are remapped to 8×8 block size to maintain consistency in comparison with ESME block size of 8×8 . Although 8×8 blocks can be further split to 8×4 , 4×8 , and 4×4 , we do not perform further splitting due to lower classification accuracy for the proposed characterization scheme as our investigations on influence of block size reveal. Please refer to Section VI-E for details.

TABLE IV
CLASSIFIER EVALUATION

Level	Classifier	Classifier mapping	Average test error (%)	
			ESME motion vectors	H.264 motion vectors
L_1	nn_1	w_1	1.73	2.21
L_2	nn_2	w_2	6.64	7.87
L_3	nn_3	w_3	0.20	0.45

B. Learning Classifiers

A total of 30 000 frames are manually labeled and features, as described in Section II-C1, are extracted to carry out the supervised learning at each level of the hierarchical scheme. Each class has 5000 samples corresponding to a particular camera pattern type. Backpropagation trained feed-forward neural network classifier is used to learn the decision boundary at various levels of the hierarchical scheme. PRTTools function `bpxnc()` is used for this purpose which uses a log-sigmoid activation function. Since the decision is made at three levels, i.e., L_1 , L_2 , and L_3 , three neural network classifiers nn_1 , nn_2 , and nn_3 are trained to obtain the mappings, i.e., w_1 , w_2 , and w_3 , respectively. These mappings are used by the hierarchical scheme for classification. The feature data corresponding to the training set is available on our webpage.

C. Testing the Learnt Classifiers

Cross-validation using four folds (n folds) is used for performance evaluation of the classifiers. The dataset is randomly permuted and divided into four equally sized parts using a stratified procedure. The classifier is trained on three parts (n folds-1) and the remaining part is used for testing. This is rotated over all parts. The resulting weighted average error is shown in Table IV. It is observed that L_1 and L_3 errors are small as only a two class decision has to be made at these two levels, i.e., at L_1 decision is needed between zoom versus others whereas at L_3 decision is needed between PRC versus SC. On the other hand, L_2 error is relatively high, as four classes have to be identified, i.e., PLC, TUC, TDC, and others, and the others class in turn has two more labels, thereby cumulating in the L_2 error value.

D. Quantitative Evaluation

Quantitative evaluation for the proposed characterization scheme is carried out in two ways: 1) by using confusion matrix and 2) by measuring sensitivity and false alarm rate (FAR) for real sequences. Confusion matrix is obtained by choosing 100 samples from each class, and then estimating the class labels obtained by the proposed classification scheme. These 100 samples from each category are selected by performing cross validation and taking the average, so that the efficiency of the classifier can be tested. These samples have not been used during the training stage. Tables V and VI show the confusion matrix for the proposed scheme utilizing the ESME-based block MVs and H.264 obtained MVs, respectively. It is observed from the confusion matrix in both

TABLE V
CONFUSION MATRIX FOR ESME MOTION VECTORS

True Label	Estimated Label						Totals
	Zoom	Still	Pan Left	Pan Right	Tilt Up	Tilt Down	
Zoom	97	0	0	3	0	0	100
Still	0	100	0	0	0	0	100
Pan Left	4	0	96	0	0	0	100
Pan Right	2	2	0	96	0	0	100
Tilt Up	0	0	0	1	99	0	100
Tilt Down	2	0	0	0	0	98	100
Totals	105	102	96	100	99	98	600

TABLE VI
CONFUSION MATRIX FOR H.264 MOTION VECTORS

True Label	Estimated Label						Totals
	Zoom	Still	Pan Left	Pan Right	Tilt Up	Tilt Down	
Zoom	93	0	0	7	0	0	100
Still	0	99	0	1	0	0	100
Pan Left	7	1	92	0	0	0	100
Pan Right	4	5	0	91	0	0	100
Tilt Up	0	0	2	4	94	0	100
Tilt Down	3	0	0	2	0	95	100
Totals	107	105	94	105	94	95	600

the cases that the classification is robust and confuses only when multiple camera motion patterns like pan with zoom are present. However, the H.264 has a relatively lower performance in comparison with the ESME-based block MVs due to several approximations made, while generating the MVs by the H.264 codec scheme.

The second part of the quantitative evaluation is carried out by measuring sensitivity and FAR for real sequences. Comparative analysis is carried out with Duan *et al.*'s method [16]. *Stefan* and *Desert* sequences are used to test the proposed scheme as these two sequences contain all the camera motion types. These two sequences were not used in the training stage. The manually labeled frames from these sequences serve as the ground truth. With reference to the ground truth, the correctly identified camera patterns by the proposed scheme are counted as true positive (TP) and the wrongly identified camera patterns are counted as false negative (FN). Using these counts, the following measures are calculated:

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (6)$$

$$\text{FAR} = \frac{\text{FN}}{\text{TP} + \text{FN}} \quad (7)$$

Table VII shows the sensitivity and FAR for the camera motion patterns for the two sequences in case of ESME-based block MVs as well as H.264 obtained MVs. As observed the proposed scheme is able to achieve high sensitivity and low FAR values and is able to clearly recognize the different camera motion patterns in comparison with Duan *et al.*'s method [16] for both ESME as well as H.264 MV types. This is due to the fact that the proposed scheme uses discriminative features from polar angle as well as magnitude

TABLE VII
SENSITIVITY AND FAR

Camera Pattern	ESME motion vectors				H.264 motion vectors			
	Sensitivity		FAR		Sensitivity		FAR	
	Duan's method [16]	proposed scheme	Duan's method [16]	proposed scheme	Duan's method [16]	proposed scheme	Duan's method [16]	proposed scheme
ZC	0.8035	0.98214	0.1964	0.0178	0.7681	0.9671	0.2319	0.0329
PLC	0.7747	0.98198	0.2252	0.01802	0.7432	0.9542	0.2568	0.0458
PRC	0.9243	0.9561	0.075	0.0439	0.8714	0.9317	0.1286	0.0683
TUC	0.8437	0.9477	0.1562	0.0523	0.8165	0.9265	0.1835	0.0735
TDC	0.7561	0.9412	0.2439	0.0588	0.7411	0.9179	0.2589	0.0821
SC	0.7014	0.97014	0.2985	0.02986	0.6823	0.9431	0.3177	0.0569

histograms which can clearly distinguish the six motion types in comparison with Duan *et al.*'s method [16], which is based on selecting features from the dominant clusters identified by the mean shift procedure. Moreover, the features selected by Duan *et al.*'s method [16] do not reflect the actual mutual relationship of MVs and is based on choosing dominant clusters and its associated MVs, whereas the proposed scheme exploits this mutual relationship using discriminative features like coefficient of variation, dominant angle bin placement, and count of first bin in the magnitude histogram. The excel file corresponding to the two sequences along with the measured parameters for Duan *et al.*'s method [16] as well as for the proposed scheme is available on our webpage. The proposed scheme uses a feature vector size of dimension = 3, i.e., $\{cv_s, \text{dominant_angle_bin_placement, count_magnitude_histogram_bin_1}\}$, in comparison with Duan *et al.*'s method [16], where a very large feature vector is used, i.e., $\{C_1, R_1, \dots, C_N, R_N, H_1, V_1, \dots, H_N, V_N\}$, where N is the number of clusters returned using mean shift procedure on the block MVs, C_i denotes the average value of MVs associated with cluster i and R_i denotes the normalized cluster size, i.e., $\sum_{i=1}^m R_i = 1$, H_i and V_i are the histograms of the projected positions of the MV on the horizontal and vertical axis, respectively. For each cluster, a pair of histograms H and V , containing k and l bins, respectively, are obtained. The dimension of the above feature vector in our reimplementation of Duan *et al.*'s method [16] is $p \times N + (k + l) \times N = 2 \times 4 + (10 + 12) \times 4 = 96$. The training time required for such a large feature dimension is close to 80 min compared with the proposed scheme which takes 4 min due to a small feature dimension which is clearly an advantage the proposed scheme offers over Duan *et al.*'s method [16]. Six SVMs are used by Duan *et al.*'s method [16] to classify the six camera motion patterns, whereas the proposed scheme utilizes only three neural network classifiers to learn the decision boundary at three levels of the hierarchical scheme.

E. Influence of Block Size on Classification Accuracy

In this section we investigate the influence of block size on the camera motion characterization accuracy. Block sizes of 4×4 and 8×8 are chosen for this purpose. The proposed characterization scheme is trained separately using 4×4 and

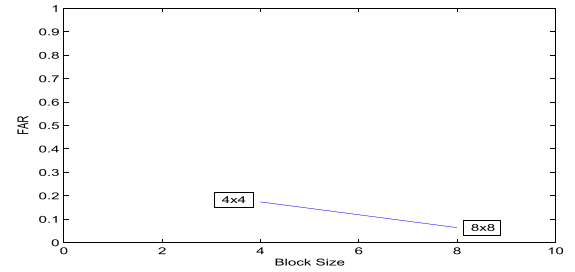


Fig. 6. Plot of average FAR versus block size for *Stefan* sequence.

8×8 block sized MVs and tested using the *Stefan* sequence. Fig. 6 shows the plot of average FAR versus the block size for the two block size cases. As observed the FAR is large for the 4×4 case as compared with 8×8 indicating lower classification accuracy for the 4×4 case. This happens due to smaller block sizes capturing outliers like object motion and noise and thereby reducing the accuracy of the camera motion classifier. On the other hand, lower FAR values are observed for larger block size (i.e., 8×8) indicating that higher sized blocks are more likely to represent the background and hence the camera motion as essentially camera motion is the motion of the background. These observations are in-line with findings by Arvanitidou *et al.* [28], where global motion estimation using variable block sizes was explored for object segmentation purposes, as well as by Khatonabadi and Bajic [29] where again, only the large blocks, i.e., 8×8 and above were used for the camera motion estimation problem while 4×4 blocks were discarded from the estimation process due to low reliability of these blocks in representing the camera motion.

F. Subjective Evaluation

The camera patterns classified by the proposed scheme are marked with colors for ease to the viewer in distinguishing the camera motion patterns. Six different colors on the border of the frames are used to distinguish the six types of camera motions identified by the proposed characterization scheme. *Stefan* and *Desert* were the test sequences used for this purpose. Table VIII lists out the coloring strategy used for the six camera motion types. As observed from the videos, the recognized camera motion patterns by the proposed scheme closely match the actual camera motions. This confirms the

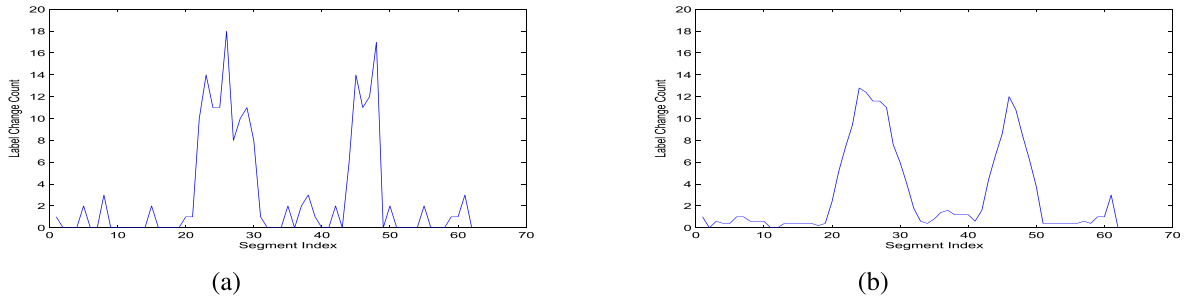


Fig. 7. Plot of label change count versus the segment index for sequence *Traffic Junction*. (a) No temporal smoothness enforced. (b) Temporal smoothness enforced using moving average filter.

TABLE VIII
COLORING STRATEGY FOR CAMERA PATTERNS

Camera Patterns	Color used
ZC	Red
PLC	Blue
PRC	Green
TUC	Yellow
TDC	White
SC	Brown

subjective performance of the proposed classification scheme with respect to the ground truth camera motion. These two marked sequences along with the excel file corresponding to the sequences giving the patterns identified with respect to the frame number are also available on our webpage.

G. Video Stabilization Application Results

This section outlines the novel application of the proposed characterization scheme to the video stabilization problem. Sequences having a combination of smooth and jittery motions are captured using a camcorder at the following locations of the Indian Institute of Technology (IIT) Kharagpur campus: 1) *Traffic Junction* near a restaurant; 2) *Scholars Avenue* (SA); and 3) *Sports Ground* (inter-IIT). The frame rate of our captured sequences is 30 frames/s. We also utilize sequences obtained from [5], [15], and [30]. Their test sequences are available at http://web.cecs.pdx.edu/~fliu/project/subspace_stabilization/, <http://web.cecs.pdx.edu/~fliu/>, and <http://www.dmi.unict.it/iplab/download/video-stab>, respectively. Sample frames from these sequences are shown in Fig. 8. These sequences are encoded with JM encoder version 18 [27] and block MVs extracted from the bitstream of these sequences are utilized.

1) *Identifying Undesired Segments*: The threshold for changes between inter-frame camera labels, i.e., label change count is set to five in this paper. Since the video frame rate of our captured sequences as well as Liu's sequences is 30 frames/s, the video segment size is set to 30, while for Puglisi's sequences it is set to 15 in accordance with their frame rate. The sequence *Traffic Junction* captured near a restaurant has both smooth as well as jittery camera motions along with moving vehicles. Fig. 7(a) shows the plot of change in label type versus the segment index for sequence

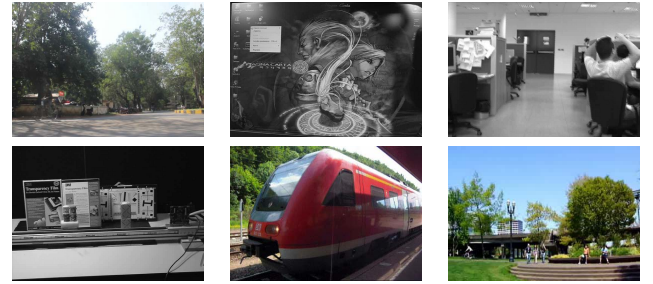


Fig. 8. Sample frames from sequences used for stabilization. First row (left to right): *Traffic Junction*, *movMouse*, and *Rotelleri*. Second row (left to right): *LuxZoom*, *Fast_Train*, and *DSCF1948*.

TABLE IX
IDENTIFIED VIDEO SEGMENTS FOR STABILIZATION

Sequence Name	Total Frames	Total Segments	Identified video segments for stabilization
<i>Traffic_Junction</i>	1830	61	21 to 29, 43 to 47
SA_1	600	20	5 to 11, 13
SA_2	2190	73	6 to 12, 19 to 22, 24, 26 to 28, 41 to 42, 57, 60 to 65
SA_3	540	18	5 to 10
SA_4	600	20	3 to 10
Inter-IIT	600	20	1 to 20 i.e. all segments
DSCF1948	510	17	1, 13 to 15, 17
movMouse	270	18	1, 2, 6 to 18
LuxZoom	270	18	1 to 18 i.e. all segments
rotelleri	240	16	1 to 16 i.e. all segments

Traffic Junction. As observed from this plot, the short-term fluctuations are affecting the label change count at several instants along the segment index values. To avoid this, we utilize the moving average filter, as described in Section IV, with $k = 2$, i.e., filter size set to five. Fig. 7(b) shows the smoothed label change count versus segment index after applying the moving average filter which smooths these spurious label changes. It is observed from Fig. 7(b) that the label change count increases beyond the set threshold (i.e., five) from segment number 21 and remains so till segment number 29. The same observation is true for segment numbers 43–47. These segments are identified by the proposed scheme and are tabulated in Table IX. Manual verification with this sequence shows the camera experiencing jittery motion during these

TABLE X
COMPUTATIONAL TIME COMPARISON

Sequence Name	Frame Resolution	Stabilization time (in seconds)		% savings obtained	Overhead
		without characterization	using proposed characterization scheme		
Traffic_Junction	1280 x 720	1870.93	702.66	62.44	No
SA_1	1280 x 720	673.45	381.86	43.29	No
SA_2	1280 x 720	2221.89	1052.12	52.64	No
SA_3	1280 x 720	554.64	285.78	48.47	No
SA_4	1280 x 720	689.47	370.37	46.28	No
Inter-IIT	640 x 480	211.69	255.47	-20.68	Yes
DSCF1948	608 x 360	136.33	90.96	33.27	No
movMouse	672 x 512	78.42	74.49	5.01	No
LuxZoom	672 x 512	79.15	91.16	-15.17	Yes
rotelleri	672 x 512	65.06	75.95	-16.74	Yes

segments confirming our observations from Fig. 7(b). Sequence *Inter-IIT* has a contingent of students performing parade during a sports event at our university campus with hand jerks at many portions of the video. This sequence is used for testing the characterization scheme for large object motions. As observed from Table IX, all the segments are marked for stabilization inline with the video characteristic. Sequences taken from [5], [15], and [30] are also utilized for performance evaluation. Sequence *Rotelleri* consists of a shaky hand camera capturing an office environment, sequence *movMouse* consists of an indoor environment where a computer mouse is moved across the unsteady camera, while sequence *LuxZoom* captures a jerky ZC on some still objects kept on a table and sequence *DSCF1948* consists of a panning camera with occasional jerks. Table IX shows the segments selected for stabilization using the proposed scheme for the mentioned test sequences. The selected segments correspond to the actual degraded portions for all the listed video sequences.

2) *Stabilizing Identified Segments*: After identifying the video segments as described earlier, the compressed domain stabilization scheme proposed in [9], whose summarized description is given in Section III-C, is run on the identified segments. Table X shows the computation time comparison for selective stabilization by integrating the proposed characterization scheme versus running the stabilization routine on all frames. The characterization time has been accounted in the complexity analysis for all sequences tested. It is observed from Table X that using the proposed characterization scheme prior to applying the stabilization routine considerable computational effort can be minimized depending on the extent of degradations (i.e., jitters) present in the sequence. For example, in the case of sequence *Traffic Junction* only 14 out of the total 61 segments require stabilization (please refer to Table IX) thereby achieving a percentage saving of 62.44 as observed from Table X. Similar is the case with other sequences used in the experimentation, namely, *SA_1*, *SA_2*, *SA_3*, *SA_4*, *DSCF1948*, and *movMouse* with varying positive percentage savings values as observed from Table X. In most cases since unwanted degradations like hand jerks and platform vibrations are not present throughout the video and occur during the course of video capture due

to uneven roads or unstable hands, the proposed application provides computational savings as the entire video need not be stabilized. However, it is also to be noted that for sequences where degradations are present in the entire video, e.g., *Inter-IIT*, *LuxZoom*, and *Rotelleri*, the proposed application of the characterization scheme to video stabilization would pose an additional overhead as seen from the negative values (ranging from 15% to 21%) in the percentage savings column. In such cases, the proposed application would pose an overhead as indicated by the overhead column because it sends all the segments ($Q \cdot s_f = N_f$) for stabilization. Detailed theoretical computational analysis for overhead is carried out in Section V.

3) *Limitation of the Proposed Method*: A limitation of the proposed camera motion characterization scheme for video stabilization application is observed when moving objects occupy more than 60% of the frame size. This would generally occur when the camera is close up or when a large moving object approaches a SC. To demonstrate the limitation, we use the sequence *Fast_Train (DSCN1113_1)* used in [15], in which a fast train is approaching a station and the user is focusing the camera on the incoming train. This sequence has 22 segments. The proposed characterization scheme identifies segment numbers 5–9 that correspond to the moving train as jittery. This happens due to prominent object motions (in this case the moving train) eclipsing the camera motions thereby limiting the proposed method from correctly identifying the movement of the camera in such cases. The stabilized results for this sequence as well as few more sequences where object motions dominate heavily are available on our webpage. The subspace video stabilization method of [15] also fails to track the KLT feature trajectories for this sequence. Similar observations have been reported in [8] for objects when they occupy more than half of the frame size. Possible solution to this problem is triggering of an ROI extractor once we detect prominent object motion eclipsing the camera motion classifier block.

VII. CONCLUSION

A novel camera motion characterization scheme is proposed by transforming the block MVs into a representation scheme using the polar angle and magnitude histograms.

Discriminative and novel statistical features from this representation are extracted for identifying the six different camera motion types. These features are learnt from the training data as part of learning a classifier, and hence are not hand coded. Object motion does not affect the proposed classification scheme, as they would be represented by tiny bins in the angle histogram. A comparative analysis with Duan's [16] method shows simplicity and robustness of the proposed scheme. The proposed scheme works at the frame level by classifying the inter-frame camera motion patterns; it is extended to classify the smooth and jittery video segments, and a novel application to video stabilization is explored. By bringing the knowledge of selective stabilization, it is shown that a considerable amount of computational time (up to 62%) can be saved, as only selective video segments that warrant stabilization are stabilized. This idea is novel as it looks into the stabilization problem from a different viewpoint, which was lacking in the literature. Our future work is focused on investigating whether a flag bit can be used to turn OFF the camera motion characterization block when prominent object motions (more than 60% frame size) start eclipsing and simultaneously trigger an ROI extractor, so that the dominant objects can be accounted for. This flag bit can again be turned ON when camera motions become dominant. In addition, we are also working on classifying multiple camera motion patterns, like pan with zoom, tilt with zoom, and so on, by grouping the sectors followed by applying the coefficient of variation for that group, which could be useful in sports videos for locating interesting events.

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